Quantum Machine Learning Seminars

Week 3: Quantum Gates & Quantum Circuits

Dr. Hassan Alshal

Outline

1. Encoding Data into Quantum States
2. Quantum Cloner
3. Quantum Gates
4. Quantum Oracles

Encoding Data into Quantum States

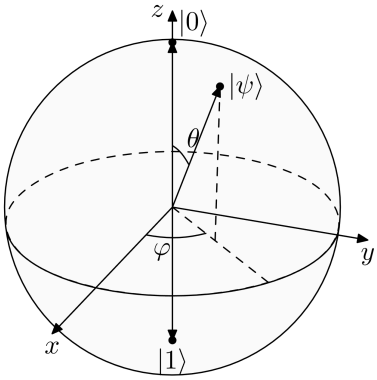
Mapping Classical Data into Quantum States

- Create an orthonormal computational basis from $x \in X \to \{|x_n^{\perp}\rangle\}_{i=1}^{n}$, e.g., $0 \to |0\rangle$ and $1 \to |1\rangle$.
- For a string $(x_1, \cdots, x_n)$ we define n-qubit $|x\rangle = |x_1\rangle \otimes |x_2\rangle \otimes \cdots |x_n\rangle = |x_1 x_2 \cdots x_n\rangle \in \mathtt{H} = \mathbb{C}^{2n}$.
- Register basic encoding: $|\psi\rangle = \dfrac{1}{\sqrt{2^n}} \displaystyle\sum_{x=0}^{2^n-1} |x\rangle$.

  And for n-features string, it requires n qubits.
- Angle encoding:

  $|\psi\rangle = R_y(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle$, where

  $$R_y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

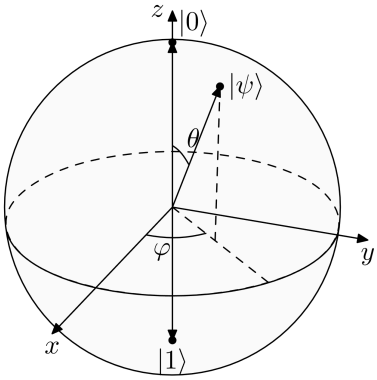


https://en.wikipedia.org/wiki/Bloch_sphere

Encoding Data into Quantum States

Mapping Classical Data into Quantum States

- Amplitude Encoding: for some feature vector $\mathbf{x} \in \mathbb{C}^n$ and some $\{\phi_i\}_{i=1}^n$, define: $|\psi_{\mathbf{x}}\rangle = \sum_{i=1}^{n} x_i |\phi_i\rangle$.
  So, for n-features, it requires $\log_2(n)$ qubits.
  e.g., $\mathbf{x} = (1,\ 2,\ 3,\ 4)$, $||\mathbf{x}|| = \sqrt{30}$ and the encoding is $|\psi_{\mathbf{x}}\rangle = \frac{1}{\sqrt{30}}|00\rangle + \frac{2}{\sqrt{30}}|01\rangle + \frac{3}{\sqrt{30}}|10\rangle + \frac{4}{\sqrt{30}}|11\rangle$.
- If the feature is a matrix $A = \sum_{i,j}^{n} a_{ij}$ with $\sum_{i,j} |a_{ij}|^2 = 1$, define $|\psi_A\rangle = \sum_{i,j} a_{ij} |\phi_i\rangle \otimes |\phi_j\rangle$.
- C.f.: Weigold, M., et al., Encoding patterns for quantum algorithms. IET Quant. Comm. 2(4), 141–152 (2021).



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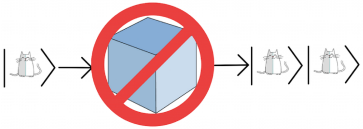
Quantum Cloner

No-cloning theorem

- Cloning means $\exists|\eta\rangle \in \mathtt{H}_B$ "blank state" and $U^\dagger = U^{-1}$ "a universal" unitary operator such that $\forall|\psi\rangle \in \mathtt{H}_A$, $U|\psi\eta\rangle = |\psi\psi\rangle$.
- There is NO such quantum cloner, i.e., superposition states cannot be copied by "reading".
- Define $U|\psi\eta\rangle = |\psi\psi\rangle$ and $U|\varphi\eta\rangle = |\varphi\varphi\rangle$.
  Then $\langle\varphi\eta|U^\dagger U|\psi\eta\rangle = \langle\varphi\varphi|\psi\psi\rangle = \langle\varphi|\psi\rangle^2$.
  At the same time $\langle\varphi\eta|U^\dagger U|\psi\eta\rangle = \langle\varphi\eta|\psi\eta\rangle = \langle\varphi|\psi\rangle$.
  Therefore, $\langle\varphi|\psi\rangle = \langle\varphi|\psi\rangle^2 \Leftrightarrow \langle\varphi|\psi\rangle \propto \delta_{ij} \Rightarrow \nexists U$.
- Try to clone $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ using $|\eta\rangle = |0\rangle$.



Quantum Cloner

No-cloning theorem

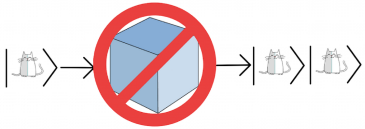
- Classical cloner is `XOR` gate: (x) and $(y) \Rightarrow (x,\ x \oplus y)$.
- The equivalent to `XOR` in the quantum context is `CNOT`: $|x\rangle|y\rangle = |x\rangle|x \oplus y\rangle$. Check how it works:

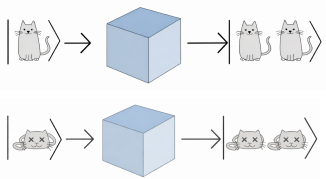
$$|0\rangle|0\rangle \mapsto |0\rangle|0\rangle \qquad , \qquad |1\rangle|0\rangle \mapsto |1\rangle|1\rangle.$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

$$(\alpha|0\rangle + \beta|1\rangle)|0\rangle \xrightarrow{\texttt{CNOT}} \alpha|0\rangle|0\rangle + \beta|1\rangle|1\rangle,$$

$$\begin{aligned}|\psi\rangle|\psi\rangle &= (\alpha|0\rangle + \beta|1\rangle)(\alpha|0\rangle + \beta|1\rangle) \\ &= \alpha^2|00\rangle + \beta^2|11\rangle + \alpha\beta|01\rangle + \beta\alpha|10\rangle.\end{aligned}$$





Quantum Gates

Quantum Registers and Quantum Gates

- An n-qubit register is system of n-qubits described as vectors in $\mathtt{H} = (\mathbb{C}^2)^{\otimes n} = \mathbb{C}^{2^n}$ that can be acted upon by n-qubit gates that are represented by $2n \times 2n$ unitary matrices.
- e.g., check how the matrix
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
acts on $|0\rangle|0\rangle$ and $|1\rangle|0\rangle$.
- Hadamrd acting on $|0\rangle$ yields basic encoding.

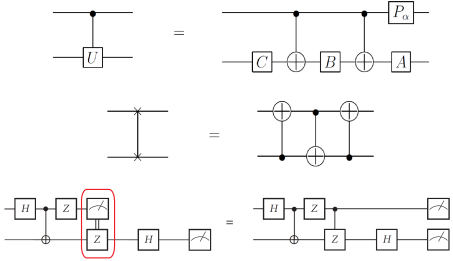
Operator	Gate(s)		Matrix
Pauli-X (X)			$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)			$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)			$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)			$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)			$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)			$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

https://en.wikipedia.org/wiki/Quantum_logic_gate

Quantum Gates

Gate Universality & Deferral

- For any 1-qubit gate U there exist unitary operators A, B, C satisfying $ABC = I$ and $\alpha \in \mathbb{R}$ such that $U = e^{i\alpha} A \sigma_x B \sigma_x C$.
- A set of quantum gates is said to be universal for quantum computation if, any n-qubit gate can be approximated to arbitrary accuracy by a composition of only those gates. $\Rightarrow \{H, S, T, \texttt{CNOT}\}$ is universal.
- Any mid-circuit computational basis measurement can be deferred to the end of the circuit, and a quantum control replaces classical-control conditioned outcome.



Quantum Oracles

Invertibility

- An oracle is defined as an abstract black box that computes $f : \mathbb{B}^n \to \mathbb{B}^m$ as a single operation.
- f isn't necessarily invertible. So, a reversible computation can be defined as the unitary operator $U_f : |x\rangle|y\rangle \to |x\rangle|y \oplus f(x)\rangle$.
- How? $|x\rangle|y\rangle \xrightarrow{U_f} |x\rangle|y \oplus f(x)\rangle \xrightarrow{U_f} |x\rangle\big|(y \oplus f(x)) \oplus f(x)\big\rangle = |x\rangle|y\rangle$, i.e., $(U_f)^1 = 1$.
- Phase Oracle: $O_f :\ |x\rangle \mapsto (-1)^{f(x)}|x\rangle$. So for example, prepare $|-\rangle = \dfrac{|0\rangle - |1\rangle}{\sqrt{2}}$.
  $U_f\big(|x\rangle|-\rangle\big) = (-1)^{f(x)}|x\rangle|-\rangle$.

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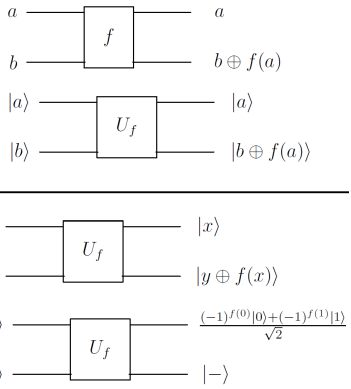

Thank You!